

WeatherSphere AI - Project Report

State-of-the-Art Meteorological Machine Learning & Web Dashboard

1. Executive Overview

WeatherSphere AI is an advanced, end-to-end meteorological predictive analytics platform that integrates time-series feature engineering, classical machine learning (Random Forest), and deep learning sequence models (PyTorch LSTM, GRU, Bi-LSTM) to perform localized weather forecasting on dynamic coordinate profiles. The system has been recently upgraded to fetch and parse live 10-day daily and 24-hour hourly Open-Meteo telemetry and display real-time Windy.com interactive satellite weather radars.

2. Core System Architecture & Data Flow

- **Data Ingestion Module:** Fetches 5 years of daily meteorological variables (temperature max/min/mean, humidity, pressure, precipitation, wind speed) from the Open-Meteo Historical Archive API.
- **Feature Engineering Engine:** Calculates short-term lag offsets (1, 3, and 7 days) and mid-term temporal rolling statistics (7-day and 30-day averages). Standardizes cyclic boundaries via Sine/Cosine conversions of month and day of year.
- **Model Zoo:** Trains an ensemble Random Forest baseline and a sequence-based PyTorch network (LSTM/GRU/Bi-LSTM) utilizing a chronological 80/20 train-test sequence split to avoid timeline leakages.
- **Streamlit GUI Client:** Displays high-fidelity analytics dashboard rendering Plotly charts, dynamic health/biometeorological alert warnings, and embedded live Windy maps.

3. Machine Learning Model Performance

Based on chronological validation tests, the deep learning PyTorch sequence model achieves a superior predictive loss margin compared to the classical Random Forest baseline. Both forecasters achieve sub-degree accuracy:

Model Architecture	Test RMSE (Celsius)	Test MAE (Celsius)
PyTorch LSTM (Active Model)	0.7014 C	0.5316 C
Random Forest (Baseline)	0.7258 C	0.5410 C

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4. Key Upgrades Implemented

- **Live Radar Integration:** Replaced static visual overlays with premium interactive Windy.com satellite weather maps centered dynamically on active coordinates, both in the main page and sidebar.
- **Actual 24-Hour Forecasts:** Replaced simulated sin-wave temperature charts with real-time hourly telemetry plotted using interactive double-axis Plotly charts.
- **WMO Code Translation:** Added a condition mapper parsing World Meteorological Organization indexes to translate raw numerical status codes into condition names and text descriptions.
- **Physiological Alert Banners:** Connected live measurements (heat load index, high gusts, barometric shifts) to dynamic biometeorological alert panels rendering medical guidelines and cardiorespiratory risk offsets.

5. Project Impact & Future Roadmap

By replacing simulated endpoints with real-time API bindings, the WeatherSphere AI platform functions as a premium diagnostic tool for meteorological exploration. The model studio enables developers and meteorologists to validate sequence-aware hyperparameter weights on local coordinate historical distributions interactively.

Planned roadmap features include supporting multi-city comparison models and anomaly alerts comparing weekly weather parameters with 30-year long-term climatological baselines.